

Management Information Systems

09. Process Modeling (II)

Guo, Xunhua

guoxh@tsinghua.edu.cn



清华经管学院
Tsinghua SEM

Process Modeling (II)

- Data Dictionaries (Textbook: Chapter 8)
- Process Specifications and Structured Decisions (Textbook: Chapter 9)



Data Dictionaries

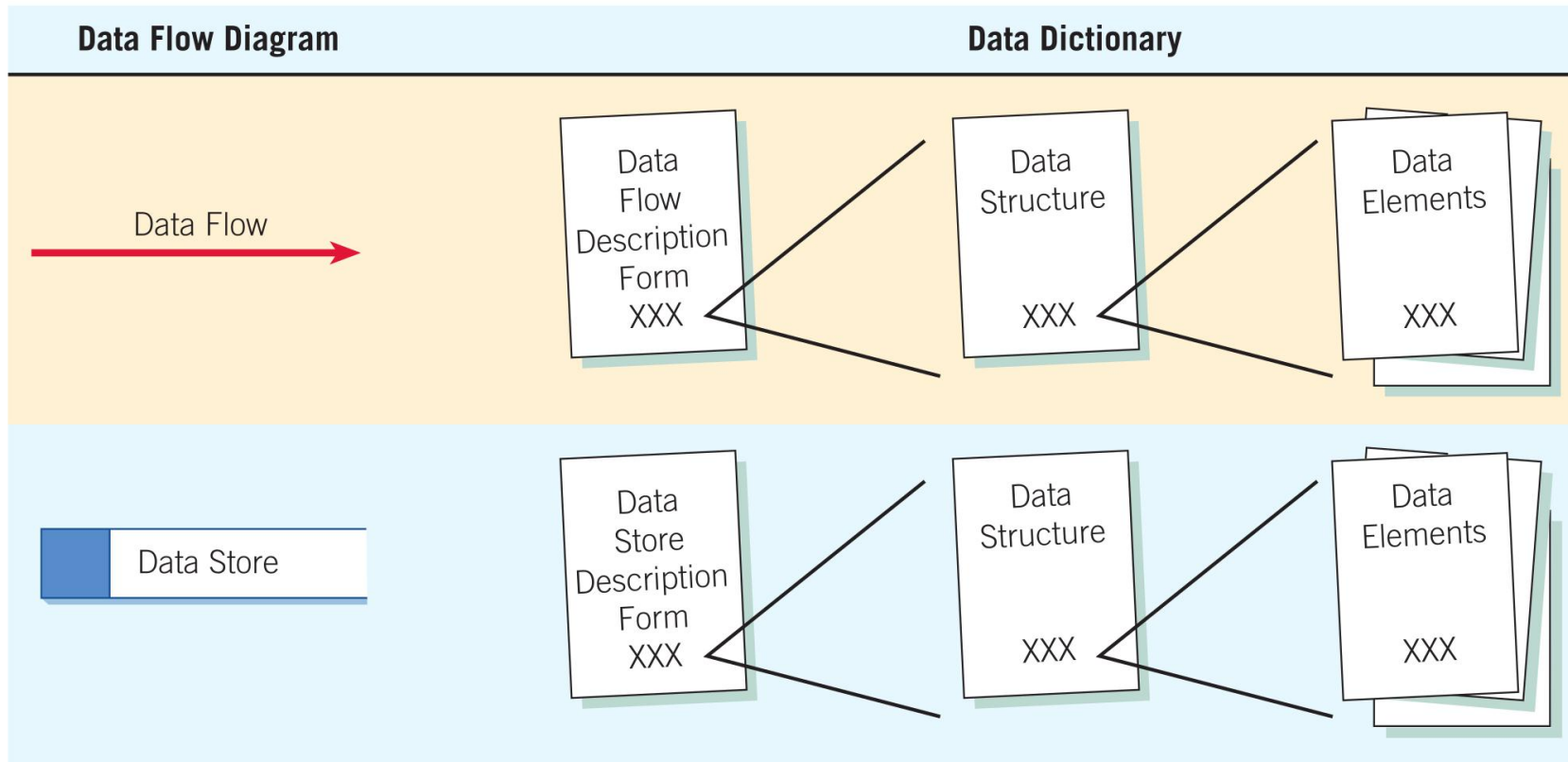


Data Dictionary

- A reference work of data about data (metadata)
- Collects and coordinates data terms, and confirms what each term means to different people in the organization



How Data Dictionaries Relate to Data Flow Diagrams



Data Dictionary Categories

- Data flows
- Data structures
- Elements
- Data stores



DFD Example

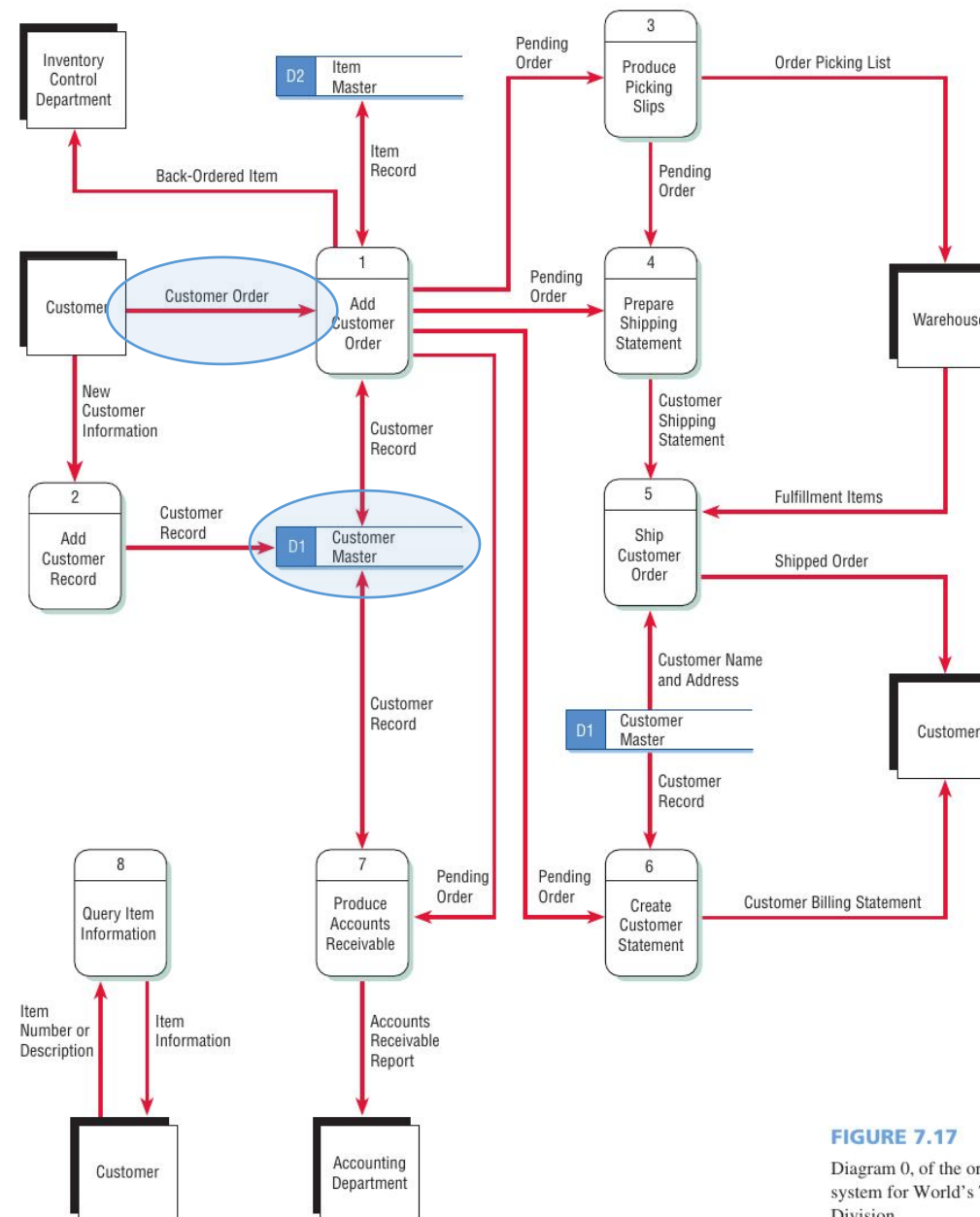


FIGURE 7.17

Diagram 0, of the order processing system for World's Trend Catalog Division.



Defining the Data Flow

- ID: identification number
- Unique descriptive name
- A general description of the data flow
- The source of the data flow
- The destination of the data flow
- Type of data flow
- The name of the data structure describing the elements
- The volume per unit time
- An area for further comments and notations

Data Flow Description	
ID _____	
Name <u>Customer Order</u>	
Description <u>Contains customer order information and is used to update the customer master and item files and to produce an order record.</u>	
Source <u>Customer</u>	Destination <u>Process 1</u>
Type of Data Flow ☐ File ☒ Screen ☐ Report ☐ Form ☐ Internal	
Data Structure Traveling with the Flow <u>Order Information</u>	Volume/Time <u>10/hour</u>
Comments <u>Order record information for one customer order. The order may be received by Web entry, email, FAX, or by the customer telephoning the order-processing department directly.</u>	

Describing Data Structures

- Data structures are made up of smaller structures and elements
- An algebraic notation is used to describe data structures
 - Equal sign = means “is composed of”
 - Plus sign + means “and”
 - Braces { } mean repetitive elements
 - Brackets [] for an either/or situation
 - Parentheses () for an optional element



Describing Data Elements

- Element ID
- The name of the element
- Aliases
- A short description of the element
- Element is base or derived
- Element length
- Type of data
- Input and output formats
- Validation criteria
- Default value
- An additional comment or remark area

Element Description Form			
ID			
Name	Customer Number		
Alias	Client Number		
Alias	Receivable Account Number		
Description	Uniquely identifies a customer who has made any business transaction within the last five years.		
Element Characteristics			
Length	6	Dec. Pt.	
Input Format	9 (6)		☐ Alphabetic
Output Format	9 (6)		☐ Alphanumeric
Default Value			☐ Date
☒ Continuous or	☐ Discrete		☒ Numeric
			☐ Base or ☒ Derived
Validation Criteria			
Continuous			
Upper Limit	<999999	Discrete Value	Meaning
Lower Limit	>0		
Comments The customer number must pass a modulus-11 check digit test. It is derived because it is computer generated and a check digit is added.			



Format Character Codes

Formatting Character	Meaning
X	May enter or display/print any character
9	Enter or display only numbers
Z	Display leading zeros as spaces
,	Insert commas into a numeric display
.	Insert a period into a numeric display
/	Insert slashes into a numeric display
-	Insert a hyphen into a numeric display
V	Indicate a decimal position (when the decimal point is not included)



Data Stores

- Data stores are created for each different data entity being stored
- Because a given data flow may only show part of the collective data that a structural record contains, many different data flow structures may need to be examined to arrive at a complete data store description

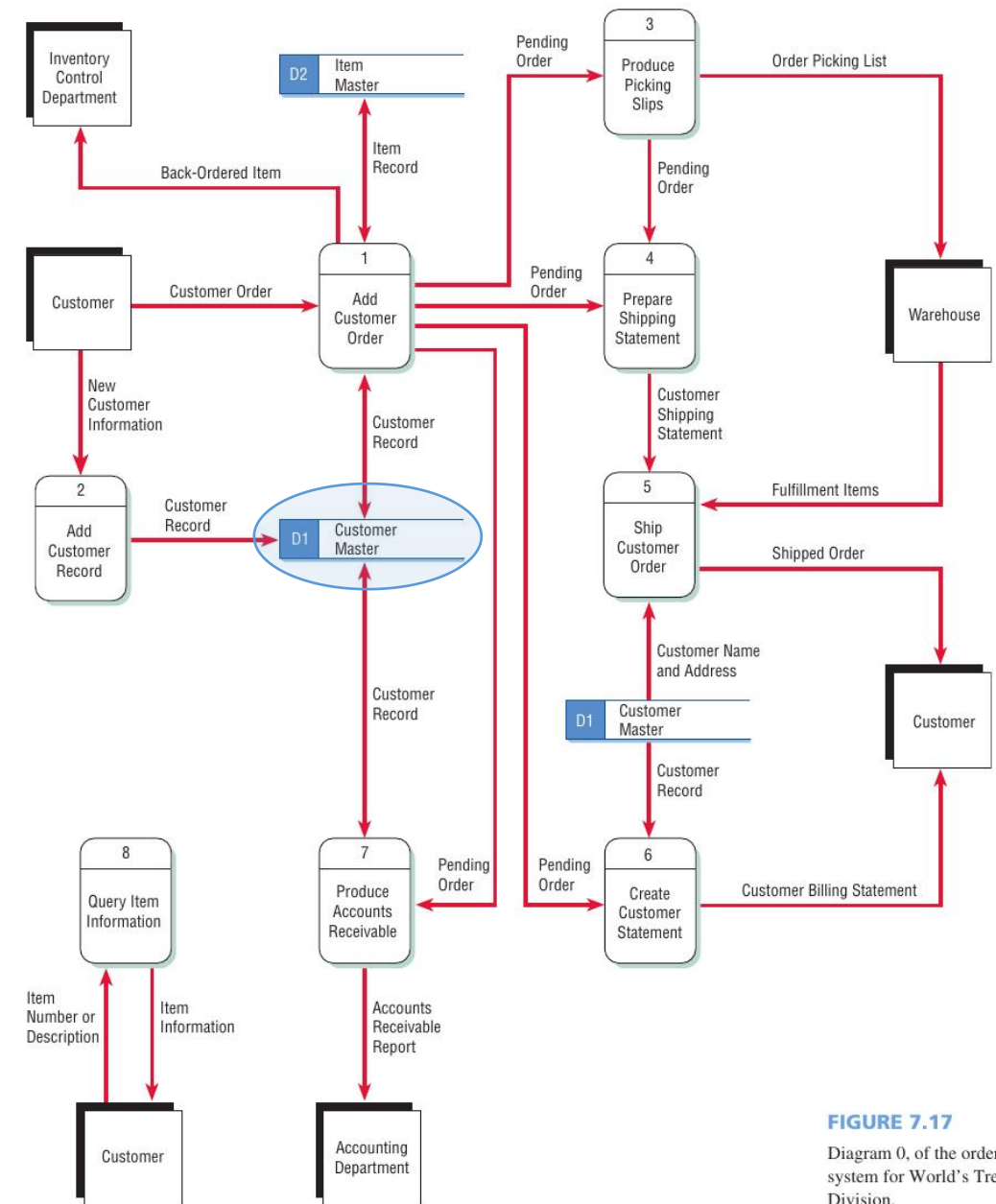


FIGURE 7.17

Diagram 0, of the order processing system for World's Trend Catalog Division.



Describing the Data Store

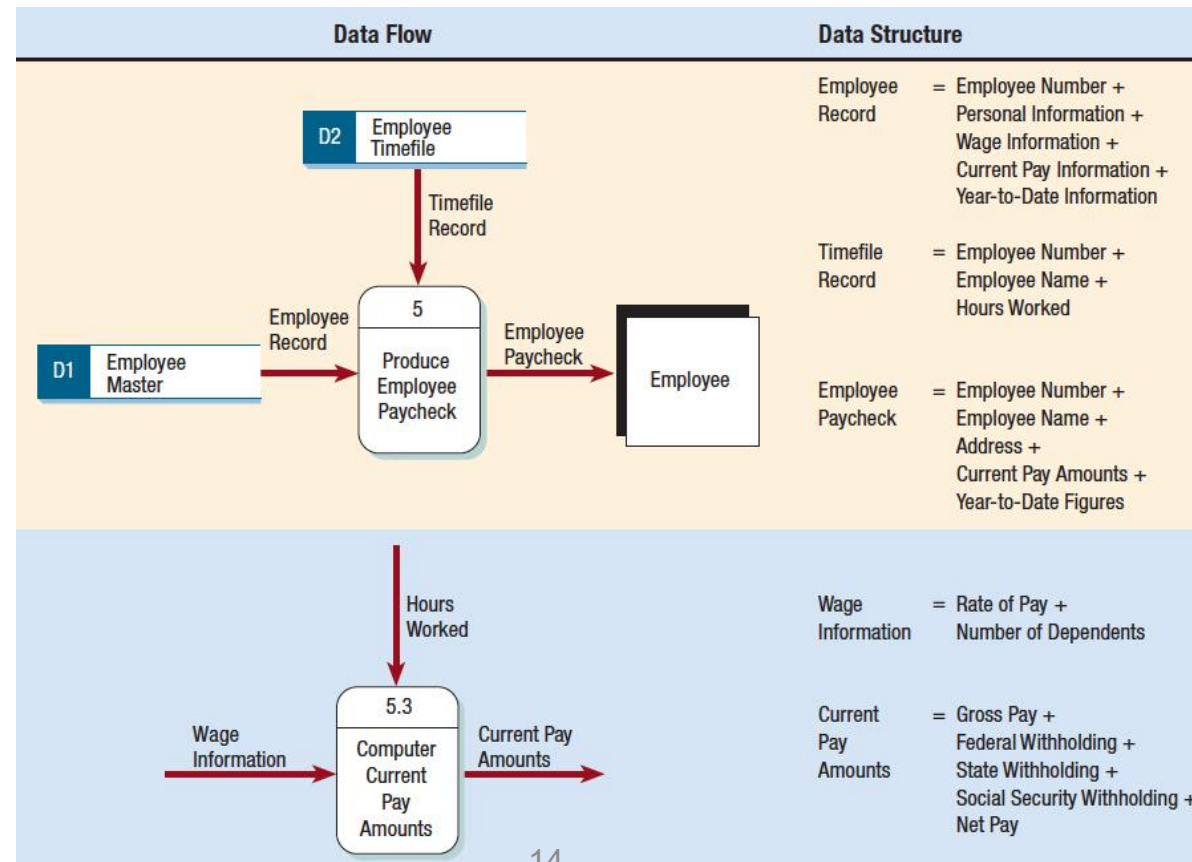
- The data store ID
- The data store name
- An alias for the table
- A short description of the data store
- The file type
- File format
- The maximum and average number of records on the file as well as the growth per year
- The file or data set name specifies the file name, if known
- The data structure should use a name found in the data dictionary
- Primary and secondary keys
- Comments

Data Store Description Form			
ID	<u>D1</u>		
Name	<u>Customer Master</u>		
Alias	<u>Client Master</u>		
Description	<u>Contains a record for each customer.</u>		
Data Store Characteristics			
File Type	☒ Computer	☐ Manual	
File Format	☒ Database	☐ Indexed	☐ Sequential ☐ Direct
Record Size (Characters):	<u>200</u>	Block Size:	<u>4000</u>
Number of Records: Maximum	<u>45,000</u>	Average:	<u>42,000</u>
Percent Growth per Year:	<u>6</u>		%
Data Set Name	<u>Customer.MST</u>		
Copy Member	<u>Custmast</u>		
Data Structure	<u>Customer Record</u>		
Primary Key	<u>Customer Number</u>		
Secondary Keys	<u>Customer Name</u>		
	<u>Zip</u>		
	<u>Year-to-Date Amount Purchased</u>		
Comments	<u>The Customer Master records are copied to a history file and</u>		
	<u>purged if the customer has not purchased an item within the past</u>		
	<u>five years. A customer may be retained even if he or she has not made a</u>		
	<u>purchase by requesting a catalog.</u>		



Creating the Data Dictionary

- Data dictionary entries are created
 - after the data flow diagram is completed or as the data flow diagram is being developed
 - using a top-down approach



Analyzing Input and Output

- A descriptive name for the input or output
- The user contact responsible
- Whether the data is input or output
- The format of the data flow
- Elements indicating the sequence of the data on a report or screen
- A list of elements

Input and Output Analysis Form

Input/Output Name Customer Billing Statement
User Contact Susan Han

File Type ☒ Output ☐ Input
File Format ☒ Report ☐ Screen ☐ Undetermined

Sequencing Element(s) Zip Code (Page Sequence)
Order Number

Element Name	Length	B/D	Edit Criteria
Current Date	6	B	(System Supplied)
Customer Number	6	D	(Includes Check Digit)
Customer First Name	20	B	Not Spaces
Customer Last Name	15	B	Not Spaces
Customer Middle Initial	1	B	A through Z or Space
Street	20	B	Not Spaces
Apartment	20	B	Not Spaces
City	20	B	Not Spaces
State	2	B	Valid State Abbr.
Zip	9	B	Numeric, Last 4 Opt.
Order Number	6	D	> 0
Order Date	8	B	MM/DD/YYYY
Order Total	9	D	Format: 9 (7) V99
Previous Payment Amount	5	D	Format: 9 (7) V99
Total Amount Owed	9	D	Format: 9 (7) V99
Comment	60	B	

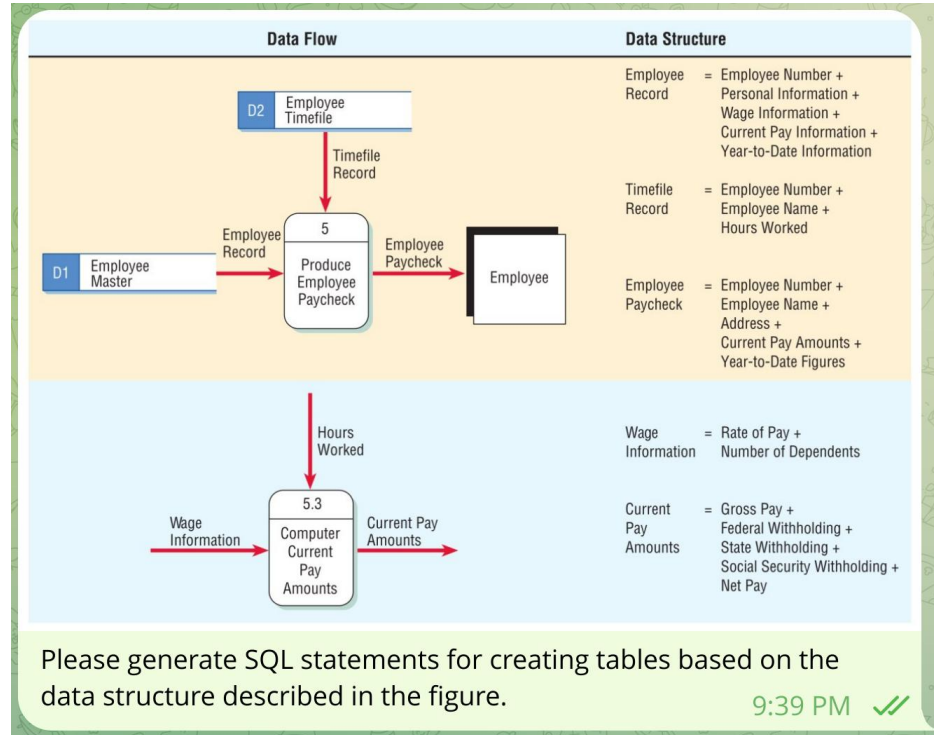
Comments Print one page for each customer. If there are more items than will fit on a page, continue on a second page.

Using the Data Dictionary

- The data dictionary may be used to
 - Create database tables
 - Create screens, reports, and forms
 - Generate computer language source code
 - Analyze the system design, detecting flaws and areas that need clarification



From the Data Dictionary to Database Tables



```
sql
-- =====
-- Payroll System Database Schema
-- Generated from Data Flow & Data Structure Diagram
-- =====

-- Table 1: employees
-- Primary data store D1: Employee Master

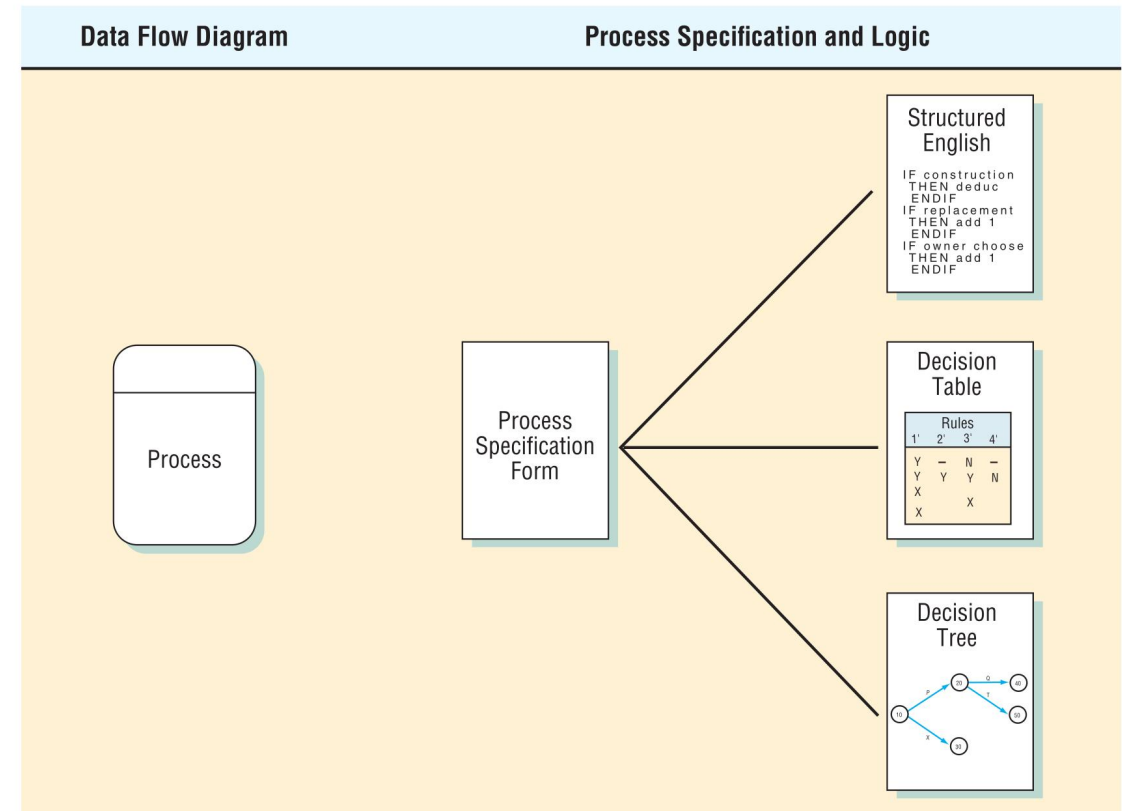
CREATE TABLE employees (
  employee_number CHAR(10) PRIMARY KEY,
  employee_name VARCHAR(100) NOT NULL,
  address VARCHAR(255),
  rate_of_pay DECIMAL(10,2) NOT NULL,
  num_dependents INTEGER DEFAULT 0,
  -- Year-to-Date fields
  ytd_gross_pay DECIMAL(12,2) DEFAULT 0.00,
  ytd_federal_tax DECIMAL(10,2) DEFAULT 0.00,
  ytd_state_tax DECIMAL(10,2) DEFAULT 0.00,
  ytd_social_sec DECIMAL(10,2) DEFAULT 0.00,
  ytd_net_pay DECIMAL(12,2) DEFAULT 0.00,
  hire_date DATE,
  department VARCHAR(50),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Process Specifications and Structured Decisions



Process Specifications (流程说明/流程规范)

- Sometimes called minispecs
- Created for primitive processes as well as for some higher level processes on a data flow diagram
- Goals
 - Reduce process ambiguity
 - Obtain a precise description of what is accomplished
 - Validate the system design



Process Specification Format

- The process number
- The process name
- Description of what the process accomplishes
- A list of input data flow
- Output data flows
- Type of process
- Uses prewritten code
- Process logic description
- Logic method reference
- List any unresolved issues

Process Specification Form	
Number <u>1.3</u>	
Name <u>Determine Quantity Available</u>	
Description <u>Determine if an item is available for sale. If it is not available, create a backordered item record. Determine the quantity available.</u>	
Input Data Flow Valid item from Process 1.2 Quantity on Hand from Item Record	
Output Data Flow Available Item (Item Number + Quantity Sold) to Processes 1.4 & 1.5 Backordered item to Inventory Control	
Type of Process ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name
Process Logic: IF the <u>Order Item Quantity</u> is greater than <u>Quantity on Hand</u> Then Move <u>Order Item Quantity</u> to <u>Available Item Quantity</u> Move <u>Order Item Number</u> to <u>Available Item Number</u> ELSE Subtract <u>Quantity on Hand</u> from <u>Order Item Quantity</u> giving <u>Quantity Backordered</u> Move <u>Quantity Backordered</u> to <u>Backordered Item Record</u> Move <u>Item Number</u> to <u>Backordered Item Record</u> DO write <u>Backordered Record</u> Move <u>Quantity on Hand</u> to <u>Available Item Quantity</u> Move <u>Order Item Number</u> to <u>Available Item Number</u> ENDIF	
Refer to: Name: _____ ☐ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: Should the amount that is on order for this item be taken into account? Would this, combined with the expected arrival date of goods on order, change how the quantity available is calculated?	



Logic Descript (1): Structured English (结构化自然语言)

- Used when the process logic involves formulas or iteration, or when structured decisions are not complex
- Based on structured logic and simple English statements such as add, multiply, and move

Structured English Type	Example
Sequential Structure A block of instructions in which no branching occurs	Action #1 Action #2 Action #3
Decision Structure Only IF a condition is true, complete the following statements; otherwise, jump to the ELSE	IF Condition A is True THEN implement Action A ELSE implement Action B ENDIF
Case Structure A special type of decision structure in which the cases are mutually exclusive (if one occurs, the others cannot)	IF Case #1 Implement Action #1 ELSE IF Case #2 Implement Action #2 ELSE IF Case #3 Implement Action #3 ELSE IF Case #4 Implement Action #4 ELSE print error ENDIF
Iteration Blocks of statements that are repeated until done	DO WHILE there are customers. Action #1 ENDDO



Advantages of Structured English

- Clarifying the logic and relationships found in human languages
- An effective communication tool, it can be taught to and understood by users in the organization



Logic Descript (2): Decision Tables (决策表)

- A table of rows and columns, separated into four quadrants:
 - Conditions
 - Condition alternatives
 - Actions to be taken
 - Rules for executing the actions

Conditions and Actions	Rules			
	1	2	3	4
Under \$50	Y	Y	N	N
Pays by check with two forms of ID	Y	N	Y	N
Uses credit card	N	Y	N	Y
Complete the sale after verifying signature.	X			
Complete the sale. No signature needed.		X		
Call supervisor for approval.			X	
Communicate electronically with bank for credit card authorization.				X



Combining Rules to Simplify the Decision Table

Conditions and Actions	Rules							
	1	2	3	4	5	6	7	8
Customer ordered from Fall catalog.	Y	Y	Y	Y	N	N	N	N
Customer ordered from Christmas catalog.	Y	Y	N	N	Y	N	N	N
Customer ordered from specialty catalog.	Y	N	Y	N	Y	N	Y	N
Send out this year's Christmas catalog.		X		X		X		X
Send out specialty catalog.			X				X	
Send out both catalogs.	X				X			

Conditions and Actions	Rules		
	1'	2'	3'
Customer ordered from Fall catalog.	—	—	—
Customer ordered from Christmas catalog.	Y	—	N
Customer ordered from specialty catalog.	Y	N	Y
Send out this year's Christmas catalog.		X	
Send out specialty catalog.			
Send out both catalogs.	X		X

Checking for Completeness and Accuracy

- Four main problems:
 - Incompleteness
 - Impossible situations
 - Contradictions
 - Redundancy



Checking the Decision Table for Impossible Situations

Conditions and Actions	Rules			
	1	2	3	4
Salary > \$50,000/year	Y	Y	N	N
Salary < \$2,000/month	Y	N	Y	N
Action 1				
Action 2				

This is an impossible situation.

Checking for Contradictions and Redundancy

Conditions and Actions	Rules						
	1	2	3	4	5	6	7
Condition 1	Y	Y	Y	Y	Y	N	N
Condition 2	Y	Y	Y	N	N	Y	N
Condition 3	—	N	—	—	—	N	Y
Action 1	X			X	X		
Action 2			X			X	
Action 3		X					X

Contradiction (indicated by arrows pointing to Rules 1, 2, and 3)

Redundancy (indicated by arrows pointing to Rules 4 and 5)

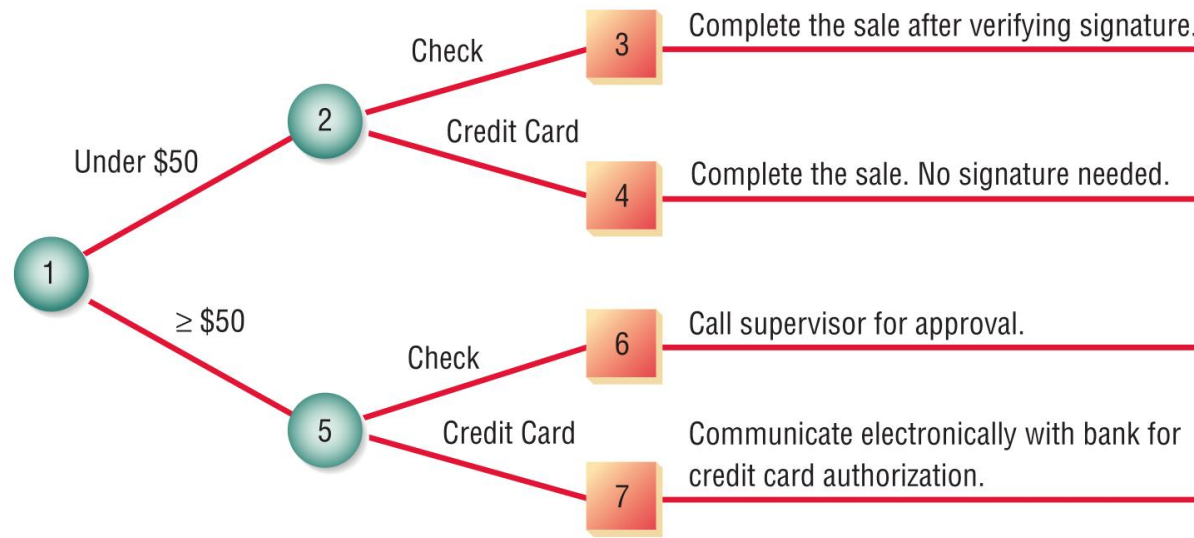
Decision Table Advantages

- Help the analysis ensure completeness
- Easy to check for possible errors
 - Impossible situations
 - Contradictions
 - Redundancy



Logic Descript (3): Decision Trees (决策树)

- Decision trees are used when complex branching occurs in a structured decision process
- Trees are also useful when it is essential to keep a string of decisions in a particular sequence
- Drawing decision trees
 - Identify all conditions and actions and their order and timing (if they are critical)
 - Begin building the tree from left to right, making sure you list all possible alternatives before moving to the right



Decision Tree Advantages

- The order of checking conditions and executing actions is immediately noticeable
- Conditions and actions of decision trees are found on some branches but not on others
- Compared to decision tables, decision trees are more readily understood by others in the organization



Selecting a Structured Decision Analysis Technique

- Use structured English when there are many repetitious actions or when communication to end users is important
- Use decision tables when a complex combination of conditions, actions, and rules are found or you require a method that effectively avoids impossible situations, redundancies, and contradictions
- Use decision trees when the sequence of conditions and actions is critical or when not every condition is relevant to every action (the branches are different)



Tasks

- Individual tasks
 - Reading: Textbook, Chapters 8~9
 - Additional review questions on WebLearning (网络学堂)
- Class presentations
 - May 11: Group 7 (蒋宇轩组, 智云科技) and Group 8 (胡布阳组, 卢米纳电子)
 - Content: requirement specifications, prototypes, process modeling
 - Time: 10~20 minutes

